

Computer Systems (ICT Skills)

Computing and Mathematics

May, 2026

Computer Systems (ICT Skills) (A13622)

Short Title: Computer Systems (ICT Skills)
Faculty Science and Computing
Department Computing and Mathematics
Cluster Networks and Cloud
Author DPOWER
Credits 5

Level: Advanced

Description of Module / Aims

The focus of this module will be on five components: computer architecture, memory management, process management, file system management and virtualisation. The student will be given hands-on experience in installing and configuring contemporary operating systems and application services with an emphasis on the Linux environment, including use of the command line interface and writing scripts.

Programmes

COMP-0518 Higher Diploma in Science in Computer Science (WD_KCOSC_G)

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Indicative Content

- Translation and calculation in the following number bases: Binary, octal, decimal, hexadecimal. Mod arithmetic
- Boolean Algebra: Basic logic & truth tables
- Computer System Architecture -- CPU, memory, storage
- Data Representation
- Operating system structure: Components, services and utilities
- Process Management: Data structures, concurrency, threads, scheduling, synchronization
- File management: File-system utilities, file system implementation, case studies
- Memory management: Paging, segmentation, virtual memory
- Scripting
- Virtualisation

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Calculate operations in binary, octal, decimal and hexadecimal number bases.
2. Apply the basics of Boolean Logic.
3. Describe how the components of a computer system operate together.
4. Demonstrate the relationship between high-level software, low-level programming and hardware.
5. Evaluate the memory, process and file management components of a modern operating system with regard to: Operation principles, data structure requirements and algorithms used.
6. Explain the concepts and theory of virtualisation and in particular how this relates to operating systems management and development.
7. Setup and configure some contemporary operating systems (within a virtual PC environment), and configure the services necessary to support basic applications.
8. Demonstrate competency with a limited set of the utilities (e.g. file management) provided by a contemporary operating system.

Learning and Teaching Methods

- This module will be delivered using a combination of lectures and practical laboratory work.
- The lectures will be used to introduce new topics and their related concepts.
- The practicals will focus on developing the practical skills of the student using simulation exercises.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,5,6
Continuous Assessment	50%	
<i>Practical</i>	25%	4,7,8
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Assessment Criteria

- <40%: Cannot represent the conceptual design of the system components presented in class or explain operation principles at a basic level.
- 40%-49%: Knows the role of each component addressed by the learning outcomes and can represent their conceptual design, supported with a basic narrative description of the operation principles.
- 50%-59%: As well as a clear understanding of the components' operation principles, can describe some of the design alternatives covered in the lectures, showing awareness for some of their strengths and weaknesses.
- 60%-69%: Can demonstrate a comprehensive understanding of the material covered in the lectures.
- 70%-100%: Excellent understanding of the presented material and displays value added knowledge as a result of independent learning.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Supplementary Material

- "Association for Computing Machinery." <http://www.acm.org>
- "Institute of Electrical and Electronics Engineers." <http://www.ieee.org>
- Dale, N. and J. Lewis. _Computer Science Illuminated_. 6th ed. MA. USA: Jones & Bartlett Learning, 2016.
- Garrido, J.M., R. Schlesinger and K. Hoganson. _Principles of Modern Operati__ng Systems_. 2nd ed. MA. USA: Jones & Bartlett Learning, 2013.
- Silberschatz, A., P.B. Galvin and G. Gagne. _Operating System Concepts with Java_. 8th ed. NJ. USA: John Wiley & Sons, 2010.
- Stallings, W. _Operating Systems: Internals and Design Principles_. 8th ed. NY. USA: Pearson, 2014.

Requested Resources

- COMPUTER LAB: BYOD Lab